SomNigil: *An Adaptive Wearable for Real-Time Sleep Distress Detection and Emergency Response*

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Most sleep-related medical emergencies go unnoticed until it is too late, particularly among elderly individuals and people living alone. By the time someone recognizes that something is wrong, valuable time may already have been lost. To address this challenge, this project presents SomNigil, a wearable system designed to continuously monitor physiological parameters during sleep and provide timely emergency alerts when potential distress is detected.

At its core, the system integrates an ESP32 microcontroller with a MAX30102 sensor to monitor heart rate and blood oxygen saturation (SpO_2), along with an MPU6050 sensor to detect body movement. Rather than relying solely on generic threshold values, the system employs an adaptive rule-based decision engine that learns each user's normal physiological patterns and continuously compares real-time sensor readings against this personalized baseline. This approach improves detection reliability while reducing unnecessary false alerts, overcoming a key limitation of conventional fitness wearables.

Upon detecting a potential distress event, the wearable immediately generates local alerts while simultaneously transmitting a timestamped notification to a companion mobile application, enabling family members or caregivers to respond promptly. Designed to be affordable, portable, and suitable for everyday home use, the proposed solution bridges the gap between consumer fitness wearables and hospital-grade monitoring systems. Beyond its technical contribution, the project supports UN SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation and Infrastructure), and SDG 11 (Sustainable Cities and Communities) by enabling safer, more independent living through timely medical intervention.